PRACTICE QUIZ

40 multiple choice · 15 true/false · 3 matching sets — his stated format. Closed book, 6:35pm.

Work it closed-book and timed. Write your answers on paper — the point is retrieval, not recognition. The answer key explains the reasoning for every item, so mark yourself honestly and go read the explanation for anything you missed *or guessed*.

Part A · Multiple Choice

1. Blood is classified as a connective tissue because:

A) It has a matrix with cells suspended in it
B) It is produced inside bone
C) It contains collagen fibres
D) It connects organs to one another

2. The normal pH range of blood is:

A) 7.25 – 7.35
B) 7.35 – 7.45
C) 7.45 – 7.55
D) 7.00 – 7.20

3. Blood viscosity is determined primarily by:

A) Plasma protein concentration
B) White blood cell count
C) The number of red blood cells
D) Water content of plasma

4. After centrifugation, plasma represents approximately what fraction of blood volume?

A) 45%
B) 90%
C) 70%
D) 55%

5. The dominant protein in plasma is:

A) Albumin
B) Thrombin
C) Fibrinogen
D) Gamma globulin

6. The primary function of albumin is to:

A) Transport oxygen
B) Maintain osmotic pressure
C) Act as an antibody
D) Form the clot mesh

7. Most plasma proteins are manufactured by the:

A) Red bone marrow
B) Kidneys
C) Liver
D) Spleen

8. The swollen abdomen seen in Kwashiorkor is caused by:

A) Enlarged liver
B) Intestinal parasites
C) Excess fat deposition
D) Fluid leaving the circulation into the tissue

9. Mature erythrocytes lack all of the following EXCEPT:

A) Hemoglobin
B) A nucleus
C) Mitochondria
D) Ribosomes

10. The biconcave shape of the RBC is advantageous because:

A) It makes the cell heavier than plasma
B) No interior point lies far from the membrane, speeding gas exchange
C) It protects the nucleus
D) It allows the cell to store more iron

11. Red blood cells generate ATP anaerobically. The functional consequence is:

A) They can survive without glucose
B) They produce ATP more efficiently than other cells
C) They do not consume the oxygen they carry
D) They divide more rapidly

12. Oxygen binds to which part of the hemoglobin molecule?

A) The globin protein chains
B) The ribosomes
C) The cell membrane
D) The iron atom at the centre of each heme

13. One red blood cell can carry approximately how many oxygen molecules?

A) 1 billion
B) 250 million
C) 4
D) 250 thousand

14. The average lifespan of an erythrocyte is:

A) 30–40 days
B) 100–120 days
C) 1 year
D) 60–80 days

15. Aged and damaged red blood cells are removed primarily by the:

A) Kidneys
B) Liver
C) Spleen
D) Red bone marrow

16. Erythropoietin is secreted by the ______ in response to ______.

A) Red bone marrow ... blood loss
B) Kidneys ... hyperoxia
C) Liver ... hypoxia
D) Kidneys ... hypoxia

17. A reticulocyte is best described as:

A) An immature RBC that has ejected its nucleus but retains ribosomes
B) A white blood cell precursor
C) An RBC that has lost its hemoglobin
D) A fragment of a megakaryocyte

18. A patient's reticulocyte count is 10%. This most likely indicates:

A) A primary bone marrow disease
B) The bone marrow is compensating for a problem elsewhere
C) Normal healthy variation
D) Vitamin B12 toxicity

19. Hematopoiesis occurs in:

A) Yellow bone marrow

B) The liver in adults
C) Red bone marrow
D) The spleen

20. Which is NOT a typical site of adult hematopoiesis?

A) Proximal femur
B) Pelvic girdle
C) Vertebrae
D) Distal tibia

21. Which nutrient sits at the centre of the heme group?

A) Iron
B) Folate
C) Calcium
D) Vitamin B12

22. The most abundant white blood cell is the:

A) Monocyte
B) Neutrophil
C) Lymphocyte
D) Basophil

23. An elevated eosinophil count most strongly suggests:

A) Bone marrow failure
B) Viral infection
C) Parasitic infection or allergy
D) Bacterial infection

24. Which white blood cell develops into a macrophage in the tissues?

A) Neutrophil
B) Eosinophil
C) Basophil
D) Monocyte

25. Histamine is released primarily by:

A) Basophils
B) Erythrocytes
C) Monocytes
D) Neutrophils

26. The ability of a white blood cell to squeeze through a vessel wall into tissue is called:

A) Phagocytosis
B) Diapedesis
C) Hemostasis
D) Chemotaxis

27. Leukopenia is most commonly induced by:

A) High altitude
B) Stress
C) Chemotherapy
D) Bacterial infection

28. Platelets are formed from:

A) Reticulocytes
B) Monocytes
C) Lymphoblasts
D) Megakaryocytes

29. The hormone that regulates platelet production is:

A) Thrombopoietin
B) Erythropoietin
C) Nitric oxide

D) Thyroxine

30. Platelets are normally kept inactive in circulation by:

A) Heparin secreted by basophils
B) Nitric oxide secreted by endothelial cells
C) Albumin
D) Low blood calcium

31. Place the steps of hemostasis in the correct order:

A) Vascular spasm → fibrin mesh → platelet plug
B) Fibrin mesh → vascular spasm → platelet plug
C) Vascular spasm → platelet plug → fibrin mesh
D) Platelet plug → vascular spasm → fibrin mesh

32. Which conversion produces the insoluble threads that seal a wound?

A) Albumin → globulin
B) Prothrombin → thrombin
C) Heme → bilirubin
D) Fibrinogen → fibrin

33. A clot that forms and remains in an unbroken vessel is called a(n):

A) Thrombus
B) Embolus
C) Infarct
D) Hematoma

34. A patient with a known DVT is at greatest risk of pulmonary embolism when they:

A) Sleep
B) Begin walking
C) Remain seated
D) Elevate the legs

35. Low-dose aspirin prevents clots by:

A) Inhibiting vitamin K
B) Dissolving existing clots
C) Blocking platelet aggregation
D) Destroying platelets

36. Warfarin (Coumadin) works by:

A) Inhibiting thrombin directly
B) Blocking platelet aggregation
C) Increasing nitric oxide
D) Interfering with vitamin K

37. Newborns receive a vitamin K injection at birth because:

A) Their gut has not yet been colonised by the bacteria that produce it
B) It prevents jaundice
C) Breast milk lacks vitamin K entirely
D) Their liver cannot process vitamin K

38. Severe liver disease causes a bleeding tendency because the liver:

A) Filters old red blood cells
B) Manufactures clotting factors, which are proteins
C) Stores platelets
D) Produces erythropoietin

39. Secondary polycythemia at high altitude is best described as:

A) An autoimmune disease
B) A form of anemia
C) A compensatory physiological response
D) A bone marrow cancer

40. After withdrawing an acupuncture needle that produces a drop of blood, you should:

A) Leave the point uncovered
B) Press and gently massage the point
C) Apply firm massage for 30 seconds
D) Press and hold with a cotton ball

Part B · True / False

Several of these are built from misconceptions the professor corrected in class. If one feels obviously true, look twice.

1. Blood thinners work by reducing the number of red blood cells. T / F

2. Aspirin reduces the ability of platelets to aggregate without destroying them. T / F

3. Atherosclerosis is caused primarily by plaque buildup rather than by blood viscosity. T / F

4. An elevated reticulocyte count is a compensatory response rather than a disease. T / F

5. Thyroid hormone directly regulates erythropoiesis. T / F

6. Neutrophils, lymphocytes, monocytes, eosinophils and basophils are all granulocytes. T / F

7. Natural killer cells contain granules that are simply not visible under light microscopy. T / F

8. A white blood cell count alone can be used to indicate sepsis. T / F

9. A stressful day can raise a person's white blood cell count. T / F

10. Plasma proteins serve as an important fuel source for blood cells. T / F

11. Increasing nitric oxide as much as possible improves platelet function. T / F

12. The red blood cell makes ATP anaerobically and does not consume the oxygen it carries. T / F

13. A cell committed to the myeloid pathway can later become a lymphocyte. T / F

14. Lymph, like blood, is classified as a connective tissue. T / F

15. Platelets are true cells with a nucleus and full organelles. T / F

16. Vascular spasm alone is sufficient to stop bleeding from a damaged vessel. T / F

Part C · Matching

1. Match the hormone to its source and target.

Term	Your answer
Erythropoietin	________
Thrombopoietin	________

A. Kidney → red bone marrow; triggered by hypoxia
B. Liver/kidney → megakaryocytes; regulates platelet output

2. Match each drug to its mechanism.

Term	Your answer
Aspirin	________
Heparin	________
Warfarin	________

A. Blocks platelet aggregation
B. Inhibits thrombin; injectable, used in hospital
C. Interferes with vitamin K; oral, used after discharge

3. Match each anemia to its cause family.

Term	Your answer
Iron deficiency anemia	________
Aplastic anemia	________
Sickle cell anemia	________
Heavy menstruation	________

A. Blood loss

B. Production fails
C. Production fails — marrow stops making all lines
D. Destruction / defect

Scoring guide. Under 80% — read the guide's ★ boxes again before bed. 80–90% — you're in good shape; drill the ⚠ traps, since that's where the true/false marks live. Above 90% — sleep.